Power System Analysis - MECH70036 Coursework

Initialization

```
clc; clear;

define_constants;
mpopt = mpoption('pf.enforce_q_lims', 1, 'verbose', 0, 'out.all', 0);

% Storage for results
summary = struct('converged', {}, 'permissible', {}, 'v_violations', {}, ...
                 'v_values', {}, 'q_violations', {}, 'q_gen_bus', {}, ...
                 'q_values', {}, 'q_min', {}, 'q_max', {}, 'q_status', {}, ...
                 'vmin', {}, 'vmax', {}, 'vmin_bus', {}, 'vmax_bus', {}, 'losses',
{});
```

Case 1: Base Case

```
%% Case 1: Base Case
fprintf('\n\n========== CASE 1: BASE CASE ==========\n');
mpc1 = case11bus;
res1 = runpf(mpc1, mpopt);
summary(1) = check_results(res1, 'Case 1');
fprintf('\n\n');
```

Case 2: Heavy Load Scenario +375 MW Load at Bus 7

```
%% Case 2: +375 MW Load at Bus 7
fprintf('\n\n========== CASE 2: +375 MW LOAD AT BUS 7 ==========\n');
mpc2 = case11bus;
P_add = 375;
Q_add = P_add * tan(acos(0.85));
b7 = find(mpc2.bus(:, BUS_I) == 7);

fprintf('Load increase: +%.0f MW, +%.0f MVAr at bus 7\n', P_add, Q_add);
mpc2.bus(b7, PD) = mpc2.bus(b7, PD) + P_add;
mpc2.bus(b7, QD) = mpc2.bus(b7, QD) + Q_add;

res2 = runpf(mpc2, mpopt);
```

```
summary(2) = check_results(res2, 'Case 2');
fprintf('\n\n');
```

Case 3: Network Reinforcement - Parallel AC Lines on 7-8 and 8-9

```
%% Case 3: Parallel Lines
fprintf('\n\n========== CASE 3: PARALLEL LINES (7-8, 8-9) ==========\n');
mpc3 = mpc2;

idx78 = find((mpc3.branch(:,F_BUS)==7 & mpc3.branch(:,T_BUS)==8) | ...
             (mpc3.branch(:,F_BUS)==8 & mpc3.branch(:,T_BUS)==7), 1);
mpc3.branch(end+1,:) = mpc3.branch(idx78,:);

idx89 = find((mpc3.branch(:,F_BUS)==8 & mpc3.branch(:,T_BUS)==9) | ...
             (mpc3.branch(:,F_BUS)==9 & mpc3.branch(:,T_BUS)==8), 1);
mpc3.branch(end+1,:) = mpc3.branch(idx89,:);

res3 = runpf(mpc3, mpopt);
summary(3) = check_results(res3, 'Case 3');
fprintf('\n\n');
```

Case 4: LCC-HVDC Solution - 350 MW Link Between Buses 7 and 9

```
%% Case 4: LCC-HVDC
fprintf('\n\n========== CASE 4: LCC-HVDC (9→7, 350 MW) ==========\n');
mpc4 = mpc2;
Pdc = 350;
Qconv = 0.60 * Pdc;
Bsh = 0.60 * Pdc;

b7 = find(mpc4.bus(:, BUS_I) == 7);
b9 = find(mpc4.bus(:, BUS_I) == 9);

mpc4.bus(b9, PD) = mpc4.bus(b9, PD) + Pdc;
mpc4.bus(b9, QD) = mpc4.bus(b9, QD) + Qconv;
mpc4.bus(b9, BS) = mpc4.bus(b9, BS) + Bsh;

mpc4.bus(b7, PD) = mpc4.bus(b7, PD) - Pdc;
mpc4.bus(b7, QD) = mpc4.bus(b7, QD) + Qconv;
mpc4.bus(b7, BS) = mpc4.bus(b7, BS) + Bsh;

res4 = runpf(mpc4, mpopt);
summary(4) = check_results(res4, 'Case 4');
```

```
fprintf('\n\n');
```

Case 5: VSC-HVDC Solution - 350 MW Link with Voltage Control

```
%% Case 5: VSC-HVDC
fprintf('\n\n========== CASE 5: VSC-HVDC (9→7, 350 MW) ==========\n');
mpc5 = mpc2;
Pdc = 350;
Qcap = Pdc * tan(acos(0.95));

b7 = find(mpc5.bus(:, BUS_I) == 7);
b9 = find(mpc5.bus(:, BUS_I) == 9);

mpc5.bus(b9, PD) = mpc5.bus(b9, PD) + Pdc;
mpc5.bus(b7, PD) = mpc5.bus(b7, PD) - Pdc;

vsc_gen = [7, 0, 0, Qcap, -Qcap, 1.0, mpc5.baseMVA, 1, 0, 0;
           9, 0, 0, Qcap, -Qcap, 1.0, mpc5.baseMVA, 1, 0, 0];
mpc5.gen = [mpc5.gen; vsc_gen];
mpc5.bus(b7, BUS_TYPE) = PV;
mpc5.bus(b9, BUS_TYPE) = PV;

res5 = runpf(mpc5, mpopt);
summary(5) = check_results(res5, 'Case 5');
fprintf('\n\n');
```

Case 6: Optimal Shunt Compensation - Minimum Reactive Power Placement

```
%% Case 6: Optimal Compensation
fprintf('\n\n========== CASE 6: OPTIMAL COMPENSATION ==========\n');
mpc_base = mpc3;
buses = 5:11;
Q_needed = inf(size(buses));

for i = 1:length(buses)
    bus = buses(i);
    Qlo = 0; Qhi = 600;

    for iter = 1:20
        Qmid = (Qlo + Qhi)/2;
        mpc_test = mpc_base;
        mpc_test.bus(bus, BS) = mpc_test.bus(bus, BS) + Qmid;
        r = runpf(mpc_test, mpopt);

        if r.success && all(r.bus(:,VM) >= 0.95 & r.bus(:,VM) <= 1.05)
            Qhi = Qmid;
```

```
        else
            Qlo = Qmid;
        end
    end

    Q_needed(i) = Qhi;
end

[Qmin, idx] = min(Q_needed);
best_bus = buses(idx);

fprintf('Optimal bus: %d, Q needed: %.1f MVAr\n', best_bus, Qmin);

mpc6 = mpc_base;
mpc6.bus(best_bus, BS) = mpc6.bus(best_bus, BS) + Qmin;
res6 = runpf(mpc6, mpopt);
summary(6) = check_results(res6, 'Case 6');
summary(6).compensation_bus = best_bus;
summary(6).compensation_mvar = Qmin;
fprintf('\n\n');
```

Summary

```
%% Summary Table
fprintf('\n\n');
fprintf('===========================================================================
=========\n');
fprintf('                              SUMMARY TABLE\n');
fprintf('===========================================================================
=========\n');
fprintf('Case  Conv  Perm  Losses(MW)  Issue                           Notes\n');
fprintf('----  ----  ----  ----------  -------------------------------
------------------\n');

for i = 1:6
    if summary(i).converged
        conv = 'Yes';
    else
        conv = 'No';
    end

    if summary(i).permissible
        perm = 'Yes';
    else
        perm = 'No';
    end
```

```
    loss = summary(i).losses;

    issue = '';
    notes = '';

    if ~summary(i).converged
        issue = 'Power flow diverged';
        notes = 'Voltage collapse';
    elseif ~isempty(summary(i).v_violations)
        worst_bus = summary(i).v_violations(1);
        worst_v = summary(i).v_values(1);
        if worst_v < 0.95
            issue = sprintf('Undervoltage at bus %d', worst_bus);
            notes = sprintf('V=%.3fpu', worst_v);
        else
            issue = sprintf('Overvoltage at bus %d', worst_bus);
            notes = sprintf('V=%.3fpu', worst_v);
        end
    elseif ~isempty(summary(i).q_violations)
        issue = sprintf('Gen %d at Q limit', summary(i).q_violations(1));
        notes = summary(i).q_status{1};
    else
        issue = 'None';
        if i == 6
            notes = sprintf('%.0f MVAr @ bus %d', summary(i).compensation_mvar,
summary(i).compensation_bus);
        else
            notes = 'All limits respected';
        end
    end

    fprintf('%-4d  %-4s  %-4s  %10.2f  %-31s  %-18s\n', ...
        i, conv, perm, loss, issue, notes);
end

fprintf('==========================================================================
=========\n\n');

%% Helper Function
function s = check_results(r, name)
    define_constants;

    s.converged = r.success;
    s.permissible = false;
    s.v_violations = [];
    s.v_values = [];
    s.q_violations = [];
    s.q_gen_bus = [];
    s.q_values = [];
    s.q_min = [];
```

```
    s.q_max = [];
    s.q_status = {};
    s.vmin = NaN;
    s.vmax = NaN;
    s.vmin_bus = NaN;
    s.vmax_bus = NaN;
    s.losses = NaN;

    if ~r.success
        fprintf('\n%s: Failed to converge\n', name);
        fprintf('System is infeasible - cannot meet constraints\n');
        return;
    end

    V = r.bus(:, VM);
    buses = r.bus(:, BUS_I);
    [Vmin, imin] = min(V);
    [Vmax, imax] = max(V);

    s.vmin = Vmin;
    s.vmax = Vmax;
    s.vmin_bus = buses(imin);
    s.vmax_bus = buses(imax);

    fprintf('\n%s Summary:\n', name);
    fprintf('  Voltages: Vmin=%.4f@bus%d, Vmax=%.4f@bus%d\n', ...
        Vmin, buses(imin), Vmax, buses(imax));

    viol = find(V < 0.95 | V > 1.05);
    v_ok = isempty(viol);

    if ~v_ok
        s.v_violations = buses(viol)';
        s.v_values = V(viol)';
        fprintf('  Voltage violations: ');
        for k = 1:length(viol)
            fprintf('Bus %d (%.4fpu) ', buses(viol(k)), V(viol(k)));
        end
        fprintf('\n');
    else
        fprintf('  Voltage status: OK\n');
    end

    Qg = r.gen(:, QG);
    Qmax = r.gen(:, QMAX);
    Qmin = r.gen(:, QMIN);
    gen_buses = r.gen(:, GEN_BUS);

    % Only display first 4 generators (exclude VSC converters)
    num_base_gens = 4;
```

```
fprintf('\n  Generator Reactive Power:\n');
fprintf('  Gen  Bus   Q(MVAr)   Qmin      Qmax      Status\n');
fprintf('  ---  ---  --------  --------  --------  ----------\n');

for g = 1:min(num_base_gens, length(Qg))
    if abs(Qg(g) - Qmax(g)) < 1
        status = 'At Qmax';
    elseif abs(Qg(g) - Qmin(g)) < 1
        status = 'At Qmin';
    else
        status = 'OK';
    end

    fprintf('  %3d  %3d  %8.2f  %8.2f  %8.2f  %s\n', ...
        g, gen_buses(g), Qg(g), Qmin(g), Qmax(g), status);
end

% Identify generators at limits for reporting
q_at_limit = find(abs(Qg - Qmax) < 1 | abs(Qg - Qmin) < 1);

% Store Q violations for summary table (but don't affect permissibility)
if ~isempty(q_at_limit)
    s.q_violations = q_at_limit';
    s.q_gen_bus = gen_buses(q_at_limit)';
    s.q_values = Qg(q_at_limit)';
    s.q_min = Qmin(q_at_limit)';
    s.q_max = Qmax(q_at_limit)';

    for k = 1:length(q_at_limit)
        if abs(Qg(q_at_limit(k)) - Qmax(q_at_limit(k))) < 1
            s.q_status{k} = 'At Qmax';
        else
            s.q_status{k} = 'At Qmin';
        end
    end
    fprintf('  Reactive power status: Gen %d at limit\n', q_at_limit(1));
else
    fprintf('  Reactive power status: OK\n');
end

[loss] = sum(get_losses(r));
s.losses = real(loss);

% Permissibility based ONLY on voltage (Q at limits is acceptable)
s.permissible = v_ok;

fprintf('\n  System losses: %.2f MW\n', s.losses);

if s.permissible
```

```
        if ~isempty(q_at_limit)
            fprintf('  => PERMISSIBLE (voltages OK, generators at Q limits)\n');
        else
            fprintf('  => PERMISSIBLE (all limits respected)\n');
        end
    else
        fprintf('  => NOT PERMISSIBLE (voltage violations)\n');
    end
end
```

Power System Analysis - MECH70036 Coursework

Case 1: Base Case

```
========== CASE 1: BASE CASE ==========

Case 1 Summary:
  Voltages: Vmin=0.9975@bus9, Vmax=1.0301@bus5
  Voltage status: OK

  Generator Reactive Power:
  Gen  Bus   Q(MVAr)   Qmin      Qmax      Status
  ---  ---  --------  --------  --------  ----------
    1    1     -1.50   -350.00    350.00  OK
    2    2    203.20   -330.00    330.00  OK
    3    3     -1.89   -580.00    580.00  OK
    4    4    232.59   -600.00    600.00  OK
  Reactive power status: OK

  System losses: 28.15 MW
  => PERMISSIBLE (all limits respected)
```

Case 1 comment: *The base case converges successfully with all voltages within the 0.95-1.05 pu limits. All generators operate well within their reactive power limits, with minimum voltage of 0.9975 pu at bus 9. System losses are moderate at 28.15 MW. The operating condition is permissible, establishing a stable baseline for comparison with subsequent cases.*

Case 2: Heavy Load Scenario +375 MW Load at Bus 7

```
========== CASE 2: +375 MW LOAD AT BUS 7 ==========

Load increase: +375 MW, +232 MVAr at bus 7

Case 2: Failed to converge
System is infeasible - cannot meet constraints
```

Case 2 comment: *Adding 375 MW load at bus 7 causes the power flow to fail to converge, indicating the system cannot maintain steady-state operation under this heavy loading. The additional demand of 375 MW and 232 MVAr exhausts the system's capacity to maintain voltage stability. Generators reach their reactive power limits during the iterative process, leading to voltage collapse. The operating condition is not permissible without network reinforcement or additional reactive support.*

Case 3: Network Reinforcement - Parallel AC Lines on 7-8 and 8-9

```
========== CASE 3: PARALLEL LINES (7-8, 8-9) ==========

Case 3 Summary:
  Voltages: Vmin=0.8899@bus7, Vmax=1.0300@bus1
  Voltage violations: Bus 7 (0.8899pu) Bus 8 (0.9081pu)

  Generator Reactive Power:
  Gen  Bus   Q(MVAr)   Qmin      Qmax      Status
  ---  ---  --------  --------  --------  ----------
    1    1    221.92   -350.00    350.00  OK
    2    2    330.00   -330.00    330.00  At Qmax
    3    3     49.47   -580.00    580.00  OK
    4    4    599.00   -600.00    600.00  OK
  Reactive power status: Gen 2 at limit

  System losses: 81.39 MW
  => NOT PERMISSIBLE (voltage violations)
```

Case 3 comment: *Adding parallel lines on corridors 7-8 and 8-9 allows power flow convergence, but the system remains non-permissible due to severe undervoltage at buses 7 (0.8899 pu) and 8 (0.9081 pu). Although the parallel lines reduce impedance by 33% and improve power transfer capability, they do not provide reactive power support. Generator 2 operates at its maximum reactive limit (330 MVAr), confirming that the primary issue is reactive power deficiency rather than thermal constraints.*

Case 4: LCC-HVDC Solution - 350 MW Link Between Buses 7 and 9

```
========== CASE 4: LCC-HVDC (9→7, 350 MW) ==========

Case 4 Summary:
  Voltages: Vmin=0.9265@bus7, Vmax=1.0300@bus1
  Voltage violations: Bus 7 (0.9265pu)

  Generator Reactive Power:
  Gen  Bus   Q(MVAr)   Qmin      Qmax      Status
  ---  ---  --------  --------  --------  ----------
    1    1    139.06   -350.00    350.00  OK
    2    2    330.00   -330.00    330.00  At Qmax
    3    3     39.63   -580.00    580.00  OK
    4    4    494.07   -600.00    600.00  OK
  Reactive power status: Gen 2 at limit

  System losses: 55.92 MW
  => NOT PERMISSIBLE (voltage violations)
```

Case 4 comment: *The LCC-HVDC link transfers 350 MW from bus 9 to bus 7, relieving AC network loading and allowing convergence. However, bus 7 voltage remains at 0.9265 pu, still below the 0.95 pu minimum limit. Although the HVDC link includes filter capacitors providing 210 MVAr at each terminal, their voltage-*

dependent nature ($Q = B \times V^2$) means they provide less support when voltages are low. The system remains non-permissible due to persistent undervoltage despite the HVDC link.

<u>Case 5</u>: VSC-HVDC Solution - 350 MW Link with Voltage Control

```
========== CASE 5: VSC-HVDC (9→7, 350 MW) ==========

Case 5 Summary:
  Voltages: Vmin=0.9873@bus7, Vmax=1.0300@bus1
  Voltage status: OK

  Generator Reactive Power:
  Gen  Bus   Q(MVAr)   Qmin      Qmax      Status
  ---  ---  --------  --------  --------  ----------
    1    1      5.39   -350.00    350.00  OK
    2    2    314.78   -330.00    330.00  OK
    3    3     28.95   -580.00    580.00  OK
    4    4    331.13   -600.00    600.00  OK
  Reactive power status: Gen 5 at limit (VSC converter represented as Gen)

  System losses: 51.61 MW
  => PERMISSIBLE (voltages OK, generators at Q limits)
```

<u>Case 5 comment:</u> *The VSC-HVDC link demonstrates superior performance by maintaining voltage control at both converter terminals through independent reactive power regulation. All bus voltages satisfy the 0.95-1.05 pu limits, with minimum voltage of 0.9873 pu at bus 7. The VSC converters operate at their maximum reactive capability (±115 MVAr each), providing the voltage support that LCC-HVDC could not. The system is permissible, demonstrating the advantage of VSC technology over LCC for weak AC systems.*

<u>Case 6</u>: Optimal Shunt Compensation - Minimum Reactive Power Placement

```
========== CASE 6: OPTIMAL COMPENSATION ==========

Optimal bus: 7, Q needed: 164.7 MVAr

Case 6 Summary:
  Voltages: Vmin=0.9500@bus8, Vmax=1.0300@bus1
  Voltage status: OK

  Generator Reactive Power:
  Gen  Bus   Q(MVAr)   Qmin      Qmax      Status
  ---  ---  --------  --------  --------  ----------
    1    1     73.15   -350.00    350.00  OK
    2    2    330.00   -330.00    330.00  At Qmax
    3    3     43.57   -580.00    580.00  OK
    4    4    519.90   -600.00    600.00  OK
  Reactive power status: Gen 2 at limit

  System losses: 74.05 MW
  => PERMISSIBLE (voltages OK, generators at Q limits)
```

Case 6 comment: *Binary search optimization identifies bus 7 as requiring the minimum reactive compensation of 164.7 MVAr among all candidate buses. This restores all bus voltages to within limits, with minimum voltage exactly at 0.9500 pu at bus 8. Generator 2 operates at its maximum reactive limit (330 MVAr), but the system is permissible as voltage constraints are satisfied. This solution demonstrates that strategic placement of shunt compensation is more cost-effective than HVDC for this particular voltage support problem.*

Summary

```
====================================================================================
                              SUMMARY TABLE
====================================================================================
Case  Conv  Perm  Losses(MW)  Issue                           Notes
----  ----  ----  ----------  ------------------------------  ------------------
1     Yes   Yes        28.15  None                            All limits respected
2     No    No           NaN  Power flow diverged             Voltage collapse
3     Yes   No         81.39  Undervoltage at bus 7           V=0.890pu
4     Yes   No         55.92  Undervoltage at bus 7           V=0.927pu
5     Yes   Yes        51.61  Gen 5 at Q limit                At Qmax
6     Yes   Yes        74.05  Gen 2 at Q limit                At Qmax
====================================================================================
```